National University of Computer and Emerging Sciences (FAST)

Differential Equations (EE) **Session-I (spring' 2016)**

Roll No:--L15-4522-- **Section:** --EE-2--

Time allowed: 60min **Max.Marks:30**

Please read the following instructions carefully before attempting any of the questions:

1. **Attempt all questions, all questions carry equal marks.**
2. **Do not ask any question about the contents of this examination from anyone.**
   a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
   b. **If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.**
   c. **Write all steps, missing steps may lead to deduction of marks.**

Q#1) Check whether the given differential equation is homogenous or not and solve it by using any appropriate substitution.

$$\frac{dy}{dx} = \frac{3x-4y-2}{3x-4y-3}$$

Q#2) Solve the given differential equation by using integrating factor.

$$dx + \left(\frac{x}{y} - siny\right) dy = 0$$

Q#3) A 100-volt electromotive force is applied to an RC series circuit in which the resistance is 200 ohm and the capacitance is 10^{-4} farad. Find the charge $q(t)$ on the capacitor if $q(0) = 0$. Find the current $i(t)$.

National University of Computer and Emerging Sciences (FAST)

Differential Equations (EE)
(Spring' 16)

Session Exam-II

Roll No:-------------------- **Section: ----------------------**

Time allowed: 60min **Max.Marks:30**

Please read the following instructions carefully before attempting any of the questions:

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Q#1) Given that $y_1(t) = t^3$ is a solution of $t^2 y'' - 6y = 0$, t>0. Use Reduction of order Method to find a fundamental set of solutions.

Q#2) Given that $2y'' + 3y' + y = x^2 + 3\sin x$

i) Find the complementary solution.

ii) Determine the general solution.

sin x 9

BS(EE)

National University of Computer and Emerging Sciences (FAST)

Differential Equations (EE) **Final Exam. (17-5- 16)**

Roll No:-------------------- **Section: -----------------------**

Time allowed: 3hrs **Max.Marks:50**

Please read the following instructions carefully before attempting any of the questions:

1. **Attempt all questions, all questions carry equal marks.**
2. **Do not ask any question about the contents of this examination from anyone.**
   a. **If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.**
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   c. **Write all steps, missing steps may lead to deduction of marks.**

Q#1) Solve the initial value problem and determine the interval in which the solution exists.

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}, y(0) = 1$$

Q#2) Find the solution for the following differential equation.

$$t^2 \frac{d^2 y}{dt^2} - 2t\frac{dy}{dt} + 2y = t^3$$

Q#3) Solve: $(x+y)dx + x\ln x dy = 0$

Q#4) a. Find the inverse Laplace transform of the given function by using the convolution theorem.

$$H(s) = \frac{s}{\left(s^2+4\right)^2}$$

b. Find the Laplace transform of the given function without evaluating the integral.

$$f(t) = \int_0^t \sin(t-\tau)\cos\tau \, d\tau$$

Q#5)

Given that $\left(4-x^2\right)y'' + 2y = 0, \quad x_0 = 0$

a. Seek power series solutions of the given differential equation about the given point x_0; find the recurrence relation.

b. Find the first four terms in each of the two solutions y_1 and y_2.

PTO